

=

Limited Resource + Unlimited Want → Scarcity

Solve scarcity

What to Produce

How to Produce [1]

For Whom

Who makes decision? [2]

[1] Efficiency (to produce, these efficiencies are needed to know)

Technical efficiency

= Maximize output for the given amount of input

= Minimize input for the given amount of output

[Quality term is needed]

Allocative Efficiency

=

Economic Efficiency

=

=

(
/)

[2] Depend on system [3]

- Capitalist [4]

- Communist [5]

- Socialist

[3]

[4]

Adam Smith (/)

- "An Inquiry into the nature and cause of the wealth of nation"

Self Interest (but like to help other) + Free market + Small Government () => Invisible hands (Supply&Demand on Market mechanism) = Best outcome

[5]

Karl Marx

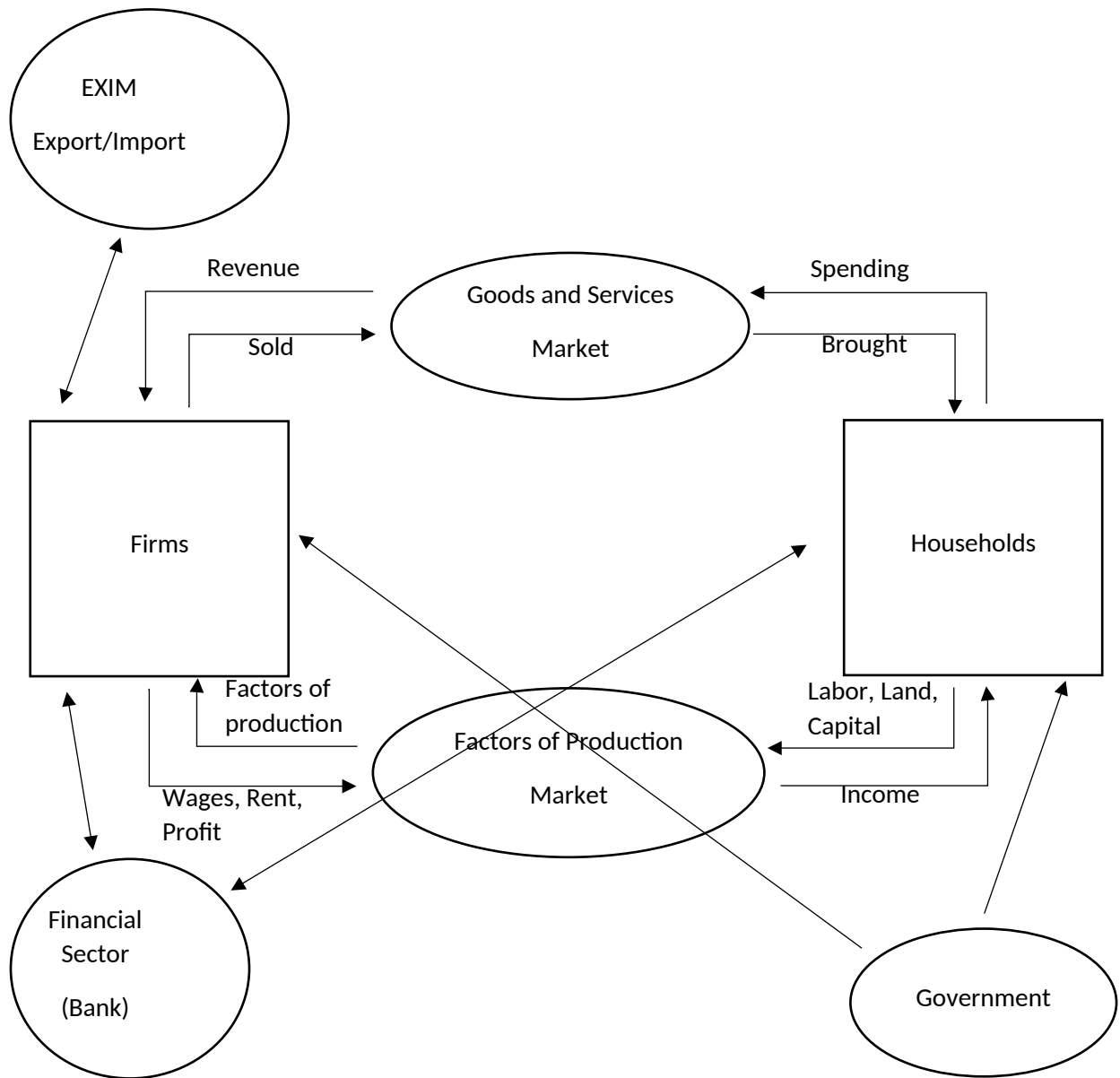
Communist manifesto

2

1. (Macro Economics) {1}

2. (Micro Economics)

{1} Macro Economic ()



4P – Marketing Mix ()

Product

Price

Place

Promotion

5C –

Character - Your credit history, reputation, and past track record of paying bills on time

Capacity - - feasibility study = Your ability to repay the loan based on cash flow, income, and debt-to-income ratios.

Capital - - The money or personal assets you have invested into the project or business as a sign of commitment

Collateral – - Assets or property you pledge to secure the loan if you cannot pay.

Condition – - The purpose of the loan and external economic factors like interest rates or industry trends

Government Goal =

1. Stable Growth -GDP&GNP [7]

2. Stable Price – [6] Creeping Inflation
(1-3%)

3. Full Employment – 4% (Natural rate of unemployment)

4. Fair Distribution of Income – [8][Gini Coefficient]
=

5. Efficiency in using resources

6. Good environment

- Aggregate Demand Money Supply(1),
(2)

2. Cost-Push Inflation

$$\frac{\text{Supply}}{\text{Demand}} = \frac{P_1}{P_2}$$

1.Demand – Pull Inflation

$$- =$$

3. Profit -Push Inflation

$$/ ()$$

$$-$$

[7]

$$\text{G.N.P./G.D.P} = \dots$$

$$\text{G.D.P. (Gross Domestic Product)} = \dots$$

$$\text{G.N.P (Gross Nation Product)} = \dots$$

$$\text{Real G.D.P.} = (\text{Nominal GDP/P.INDEX}) * 100$$

$$- \text{ Price Index (P.INDEX)} = \frac{\text{Base Year (P.INDEX = 100)}}{\text{(Best Year)}}$$

$$-$$

$$\text{C.P.I.} = \text{Consumer Price Index} =$$

$$\text{P.P.I.} = \text{Producer Price Index} =$$

$$Y = C + I + G + X - M$$

Whereas $Y = \text{G.D.P}$

$C = \text{Consumption}$

$I = \text{Investment} =$

+

$G = \text{Government Expenditure ()}$

$X = \text{Export}$

$M = \text{Import}$

According to John Maynard Keynes, $C = a + by$

Whereas $C = \text{Total Consumption}$

$a = \text{Autonomous() Consumption}$

$b = \text{Marginal Propensity to consume}$

$y = \text{Income}$

Marginal Propensity

$\text{MPC (Marginal Propensity Consume} = \text{)} = b$

$$= \frac{C}{y}; \quad 0 \leq \text{MPC} \leq 1$$

$\text{MPS (Marginal Propensity to Save} = \text{)}$

$$= \frac{S}{y}; \quad 0 \leq \text{MPS} \leq 1$$

$$\text{MPC} + \text{MPS} = 1$$

$$Y = C + I + G + X - M \quad \text{----}\{1\}$$

Case 1 No tax

$$C = a + by \quad \text{----}\{2\}$$

Combine {2} in {1};

$$Y = \frac{a + I + G + X - M}{1 - b}$$

$$Y = \frac{a + I + G + X - M}{1 - b}$$

$$\text{*Multiplier} = \frac{1}{1 - b}$$

Case 2 Lump Sum tax ($\quad \quad \quad = T$)

$$C = a + b(y - T) \quad \text{----}\{3\}$$

Combine {3} in {1};

$$Y = \frac{a - bT + I + G + X - M}{1 - b}$$

$$Y = \frac{a - bT + I + G + X - M}{1 - b}$$

Case 2 + Balance Budget Multiplier ($\Delta T = \Delta G$;) +

Combine {3} in {1} with the condition that $\Delta T = \Delta G$ and $\Delta a = \Delta I = \Delta X = \Delta M$

$$\Delta Y = \Delta G = \Delta T$$

Case 3 Flat-rate tax ($\quad \quad \quad , \quad \quad \quad = ty$)

$$C = a + b(y - ty) \quad \text{---}\{4\}$$

Combine {4} in {1};

$$Y = \frac{a + I + G + X - M}{1 - b + bt}$$

$$Y = \frac{a + I + G + X - M}{1 - b + bt}$$

[8] Gini Coefficient (GINI INDEX)



CORRADO GINI

$$\text{GINI COEFFICIENT} = \frac{A}{A+B}$$

$$= 0 \quad =$$

$$= 1 \quad =$$

$$=$$

_____ = asset = for Medium of Exchange, Unit of Account, Store of Value

- Commodity Money = have Intrinsic value

- Fiat Money

Cryptocurrency

Block Chain =

Bitcoin – Satoshi Nakamoto

Economist

+

Financial Market / Financial Intermediaries

Type – Organized market, Unorganized Money Market

System – Money market (Financial transaction < 1 yr), Capital market (Financial transaction ≥ 1 yr)

Money Supply

Narrow money (M1) - - Cash +
(Current deposit)

Broad money (M2) - - M1 +
(Saving) + (Fixed)

Broad money (M2a) - - M2 + Promissory
note (finance coms borrow public)

Broad money (M3) - - All money

(2) _____

- **Quantitative Tool** = control =>

Require Reserve [9]

Discount Rate / Federal Discount Rate [note; let Inter Bank Rate (or Federal Fund Rate) work first; Discount > Fund]

Open market operation ()

- **Qualitative Tool** = control =>

Selective Interest Control

Selective Credit Control

Moral Suasion ()

[9] Require Reserve

DD (Demand Deposit) → TR (Total Reserve) = RR (Required Reserve) + ER
(Excess Reserve)

Total Maximum Money Supply [Original Deposit/rr] = Money Supply From bank
+ Original Deposit (or Demand Deposit)

*Note: The formula for Total Max Money Supply is for 100% money return in
bank but, in reality, it is around 70%

Foreign Exchange Market (Forex, Fx, Currency Market)

International Reserves

= Foreign Currencies, Gold, Special Drawing
Right (S.D.R)

Should ≥ 6 month of Import Value

For -1- International debt settlement () -2-
-3-

Exchange Rate

- Fixed Exchange Rate - up-
Devaluation down-Revaluation

- Flexible Exchange Rate / Floating Exchange Rate - up-
Depreciation down-Appreciation

Dirty float(/Managed float)

Currency Basket = ()

Balance of Payment [BoP]

Current Account () = Trade Account, Service Account, Transfer Account

Capital Account

BoP (Balance of Payment) = NX (Net Exports) + CF (Capital/Financial Flow)

NX Depends on $Y^*(+)$ $Y(-)$ $P^*(+)$ $P(-)$ $E(-)$

CF Depends on $I(+)$ $I^*(-)$

Where as, * = Foreign

Y = income or GDP

P = Price

E = Exchange Rate

I = Interest Rate

Real Exchange Rate: RER (or E) = $e(P^*/P)$; e = Nominal Exchange Rate

Risks → Country risk, Currency risk, Credit risk

Managerial Economics = Economic Concepts + Quantitative Methods

Business Profit = Total Revenue (TR) – Total Accounting (Explicit) Cost

Economic Profit = Total Revenue (TR) – [Total Accounting (Explicit) Cost + Opportunity (Implicit) Cost]

Demand and Supply

Law of Diminishing of Return =

= While

$$\text{Marginal [Cost } \vee \text{ Price]} = \frac{\partial(\text{Total Cost } \vee \text{ Price})}{\partial(\text{Quantity})}$$

Demand

= Quantity of good/service that customers are willing and able to purchase under various market condition

Law of Demand = Negative relationship b/w Quantity demanded and Price

Direct Demand = Demand for Consumption products

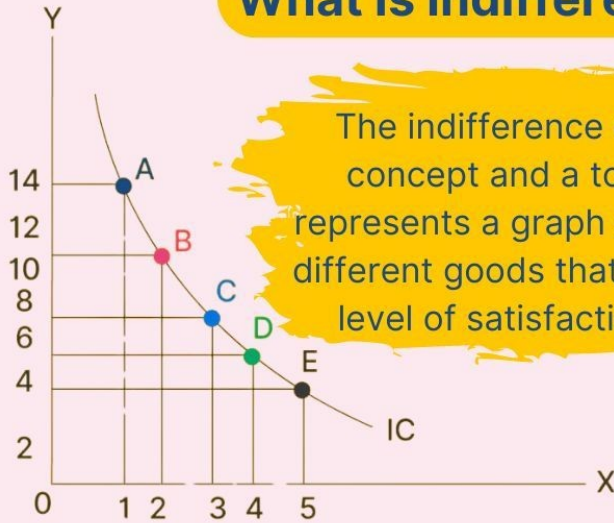
Derived Demand = Demand for Input used in production

Indifference curve =

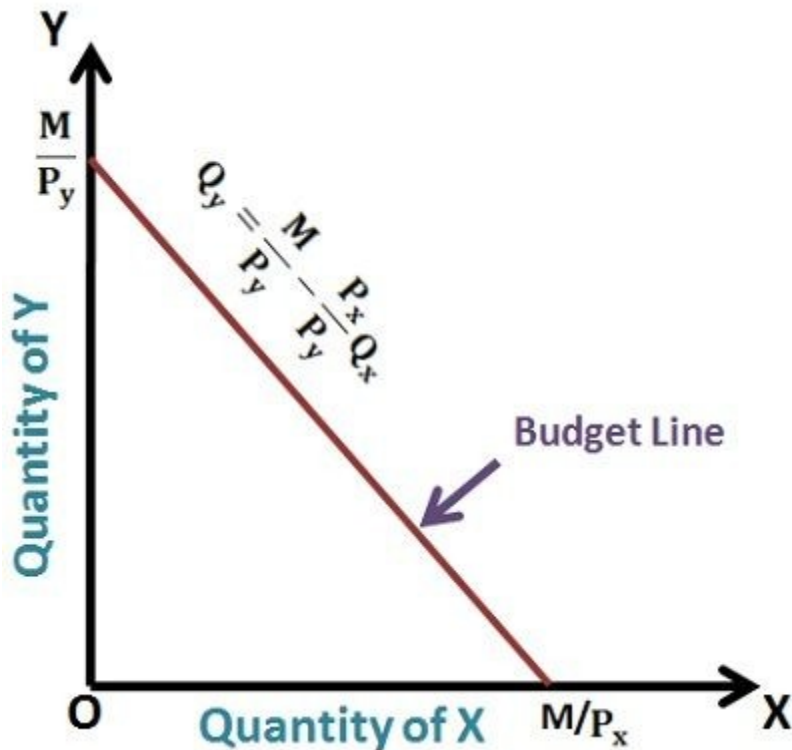
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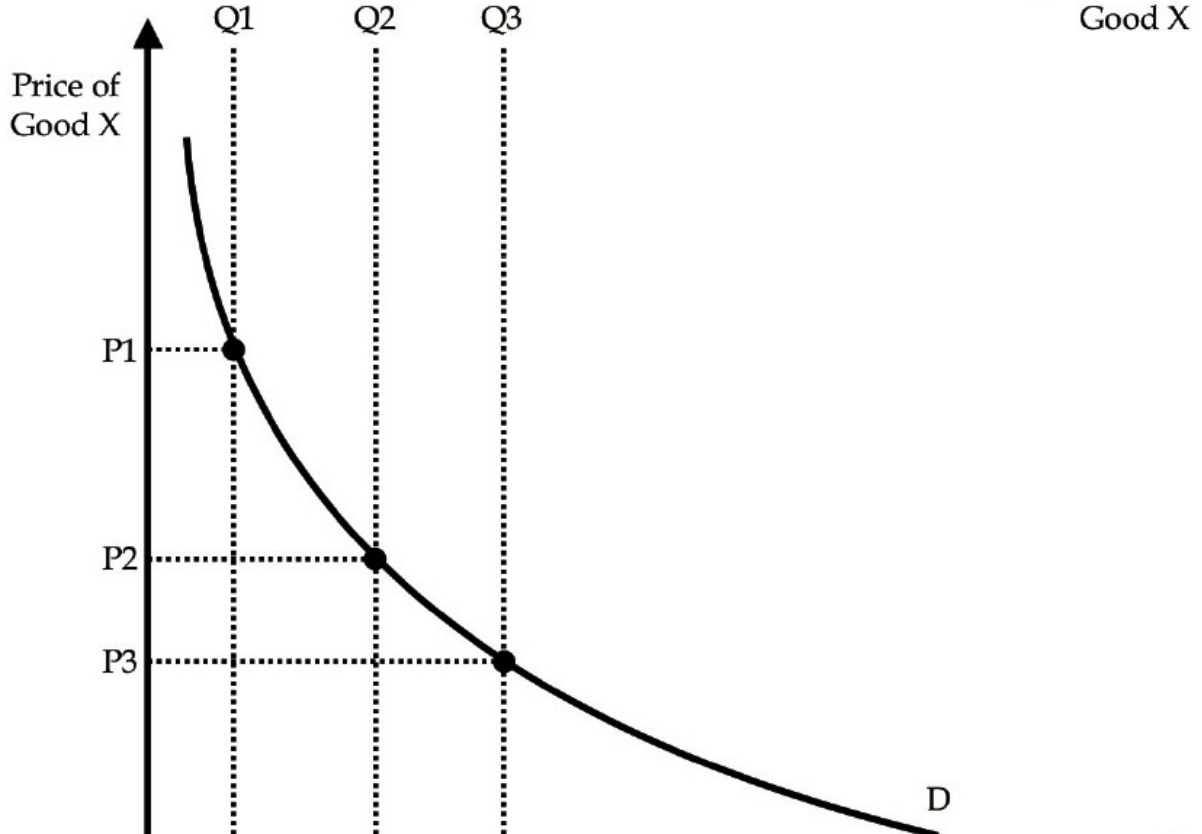
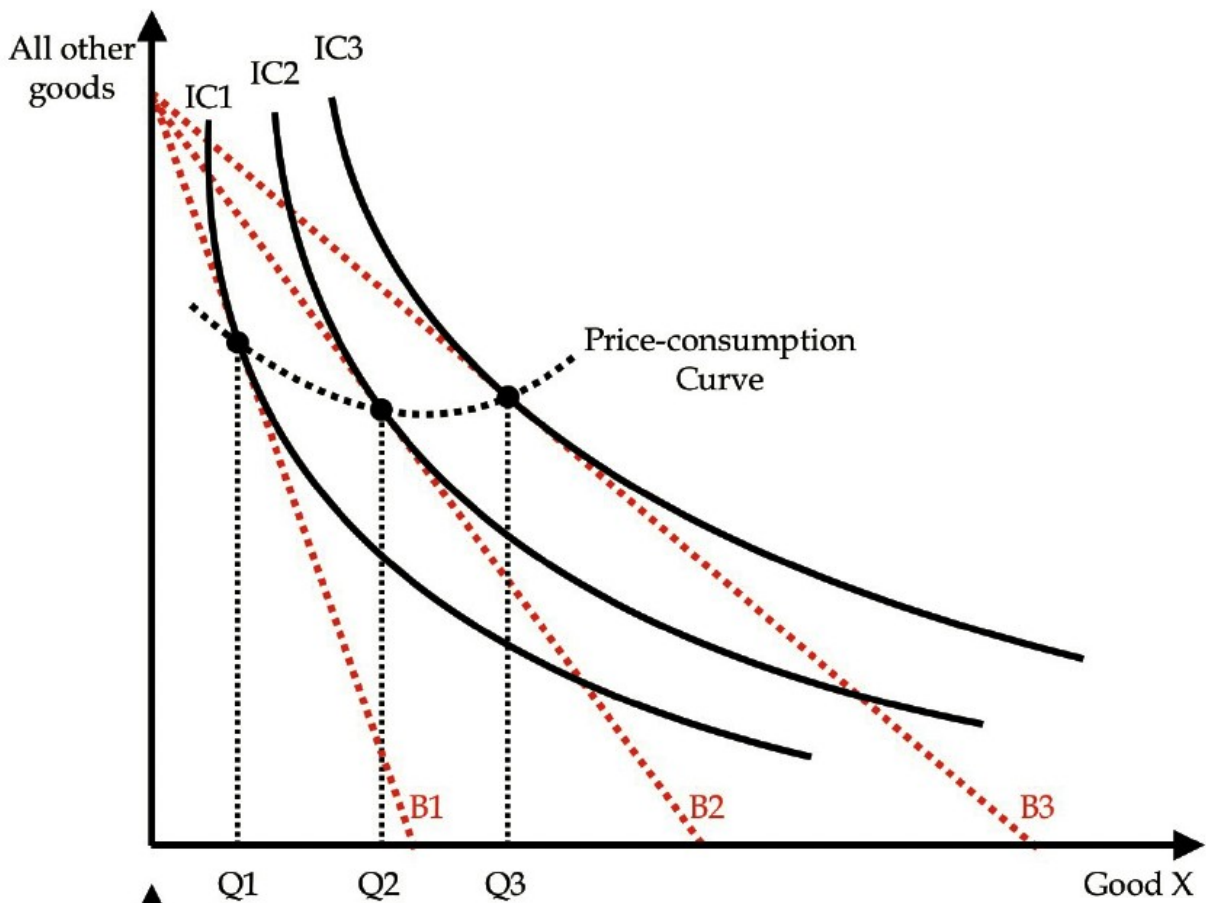
Slope

What Is Indifference Curve?



The indifference curve (IC) is a fundamental concept and a tool in microeconomics that represents a graph showing combinations of two different goods that essentially provide the same level of satisfaction or utility to a consumer.





Change in quantity demanded = Demand

Change in demand = demand []

Supply

Law of supply = Quantity of supplied of good rises when Price rises

Change in quantity supplied = Supply

Change in supply = demand []

Demand and Supply

Equilibrium = basket is clear =

Surplus = Excess supply =

Shortage = Excess demand =

Price control –

[Price Ceiling] vs

[Price Floor]

Demand Sensitivity Analysis: **Elasticity** =

$$Elasticity = \frac{\% \Delta y}{\% \Delta x}$$

y = Dependent variable

x = Independent variable

$$Point \ Price \ elasticity \ of \ demand = \frac{\% \Delta Quantity \ demand \ of \ x}{\% \Delta price \ of \ x}$$

Arc Price Elasticity of demand

5 %

$$Arc \ Price \ elasticity \ of \ demand = \frac{Q_2 - Q_1}{P_2 - P_1} \times \frac{P_2 + P_1}{Q_2 + Q_1}$$

Q = Quantity Demand of x

P = Price of x

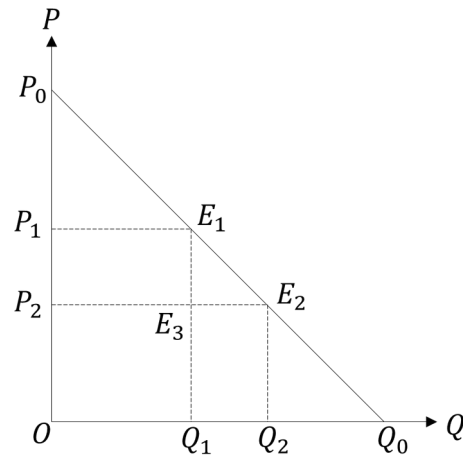
$$Arc \ Price \ elasticity \ of \ demand = \frac{\% \Delta Q_x}{Large \% \Delta P_x}$$

$|EP| < 1 \rightarrow$ Inelastic

$|EP| = 1 \rightarrow$ Unitary elastic

$|EP| > 1 \rightarrow$ Elastic

Point Price Elasticity of Demand



$$E_p = \frac{\% \cdot \Delta Q}{\% \cdot \Delta P}$$

$$= \frac{\frac{Q_2 - Q_1}{Q_1} \times 100}{\frac{P_2 - P_1}{P_1} \times 100}$$

$$= \frac{Q_2 - Q_1}{P_2 - P_1} \times \frac{P_1}{Q_1}$$

$$= \frac{\Delta Q}{\Delta P} \times \frac{P_1}{Q_1}$$

$$= \frac{\partial Q}{\partial P} \cdot \frac{P_1}{Q_1}$$

Arc Price Elasticity of Demand

$$E_p = \frac{\% \cdot \Delta Q}{\% \cdot \Delta P}$$

$$= \frac{\frac{Q_2 - Q_1}{\left(\frac{Q_1 + Q_2}{2}\right)} \times 100}{\frac{P_2 - P_1}{\left(\frac{P_1 + P_2}{2}\right)} \times 100}$$

$$= \frac{Q_2 - Q_1}{P_2 - P_1} \times \frac{P_1 + P_2}{Q_1 + Q_2}$$

$$= \frac{\Delta Q}{\Delta P} \times \frac{P_1 + P_2}{Q_1 + Q_2}$$

$$= \frac{\partial Q}{\partial P} \cdot \frac{P_1 + P_2}{Q_1 + Q_2}$$

(Price Elasticity of Demand)

$$E_p = \frac{\% \cdot \Delta Q}{\% \cdot \Delta P}$$

E↓

vs

E↑

/

ΔP	$ EP < 1$	$ EP = 1$	$ EP > 1$
P↑	TR↑	$\overline{TR}$	TR↓
P↓	TR↓	$\overline{TR}$	TR↑

E_p = Elasticity =

1

$$\pi = TR - TC$$

$$\pi = PQ - (CQ + F)$$

$|EP| < 1$

↑

P↑ TR↑ and TC↓

$|EP| > 1$

P↓ TR↑ and TC↑

Income Elasticity of Demand (E_y)

$$E_y = \frac{\% \Delta Q}{\% \Delta y}$$

Q = Quantity, y = income

$E_y > 1$ => Superior goods / Cyclical normal goods

$0 \leq E_y \leq 1$ $\Rightarrow$ Normal goods / Non-cyclical normal goods

$E_y < 0$ $\Rightarrow$ Inferior goods [ep.]

Cross-price Elasticity of Demand (E_{xy})

$$E_{xy} = \frac{\% \Delta Q_x}{\% \Delta P_y}$$

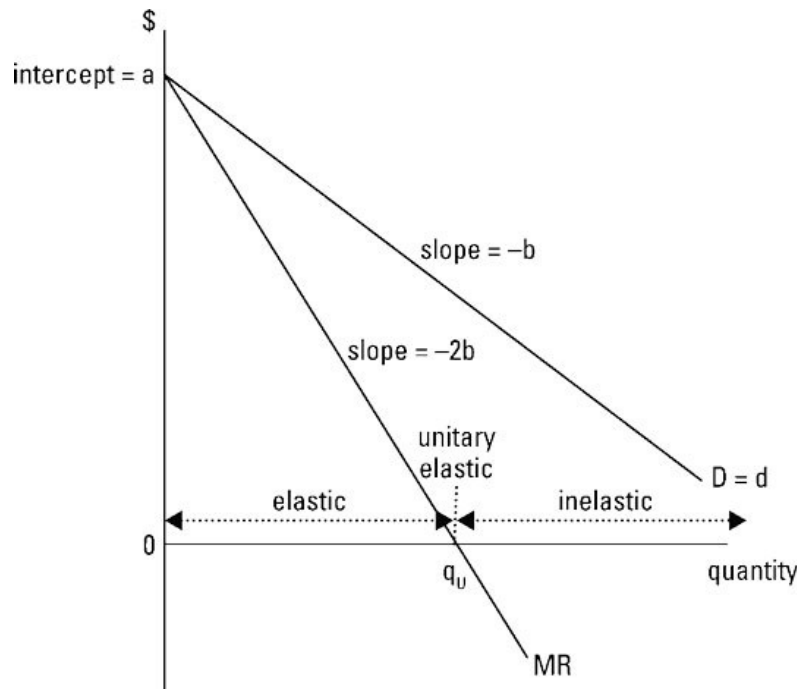
$E_{xy} < 0$ $\Rightarrow$ = Complementary goods

$E_{xy} > 0$ $\Rightarrow$ = Substitution goods

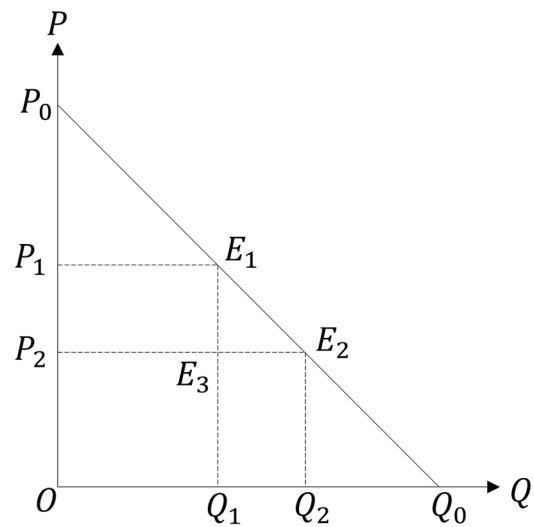
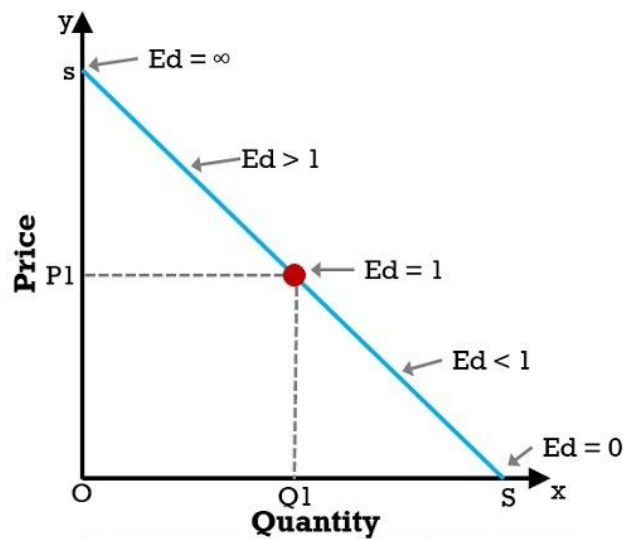
$E_{xy} = 0$ $\Rightarrow$ = Independent goods

	Demand	Marginal Revenue
$P = a - bQ$	-----1	
$TR = P \cdot Q$	-----2	
$TR = (a - bQ) \cdot Q$		
$TR = aQ - bQ^2$	-----3	
$MR = \frac{\partial TR}{\partial Q} = a - 2bQ$		

So, MR's slope = 2x Demand's slope



Point Price Elasticity of Demand @ Middle



$$E_p = \frac{\Delta Q}{\Delta P} \cdot \frac{P}{Q}$$

$$E_p = \frac{\Delta Q}{\Delta P} \times \frac{P}{Q}$$

$$E_p = \frac{Q_2 - Q_1}{P_2 - P_1} \times \frac{OP_1}{OQ_1}$$

$$E_p = \frac{E_2 E_3}{E_1 E_3} \times \frac{O P_1}{O Q_1}$$

$$E_p = \frac{Q_0 Q_1}{Q_1 E_1} \times \frac{O P_1}{O Q_1}$$

From $Q_1 E_1 = O P_1$

$$E_p = \frac{Q_0 Q_1}{O Q_1}$$

$$E_p = \frac{Q_0 E_1}{E_1 P_0}$$

E_1

Demand

$E_p = 1$

Price, TR, MR

$$TR = P \cdot Q$$

$$MR = \frac{\partial TR}{\partial Q} = P \cdot \frac{\partial Q}{\partial Q} + Q \cdot \frac{\partial P}{\partial Q}$$

$$MR = P + \frac{Q \partial P}{\partial Q}$$

$$MR = P \left(1 + \frac{Q}{P} \cdot \frac{\partial P}{\partial Q} \right)$$

$$MR = P \left(1 + \frac{1}{E_p} \right)$$

Firm Maximize Profit

MC = MR (MC Slope +)

$$MC = P \left(1 + \frac{1}{E_p} \right)$$

$$P = \frac{MC}{1 + \frac{1}{E_p}}$$

36

5%

32

?

$$\begin{aligned} &16 \\ &MC = 10 \end{aligned}$$

$$E_p = \frac{\Delta Q}{\Delta P} \cdot \frac{P}{Q}$$

$$E_p = \frac{36 - 32}{-5\% (16)} \cdot \frac{16}{32}$$

$$E_p = \frac{4}{-0.8} \cdot \frac{16}{32}$$

$$E_p = -2.5$$

$$P = \frac{MC}{1 + \frac{1}{E_p}}$$

$$P = \frac{10}{1 + \frac{1}{-2.5}}$$

$$P = 16.67$$

Demand Estimation

Consumer Interviews → Target Group, Pilot Test, Train Interview

– Cons = Not real feeling of consumer, Other uncontrol Effect

Market Experiments

Laboratory Experiment + Pros = No outside effect - Cons = Not real feeling, Can't be long time

Actual Market Experiment + Pros = Real

- Cons = Outside Effects, Confused by enemy,
Permanent loss consumers from increasing price, Can't
be long time, huge budget

Regression Analysis

$$y_t = \hat{a} + \hat{b} x_t$$

$$\hat{b} = \frac{\sum_{t=1}^n (x_t - \bar{x})(y_t - \bar{y})}{\sum_{t=1}^n (x_t - \bar{x})^2}$$

$$\hat{a} = \bar{y} - \hat{b}\bar{x}$$

BLUE = Best, Linear, Unbiased, Estimator

Error term = Normally Distributed, Expected Value = 0, Constant Variance, Independent Error

Unbiased Estimator $\rightarrow \text{Bias} = E(\hat{b}) - b$ ** Null & Alternative Hypothesis

$$S_b = \sqrt{\frac{\sum (y_t - \hat{y})^2}{(n-k) \sum (x_t - \bar{x})^2}} = \sqrt{\frac{\sum e_t^2}{(n-k) \sum (x_t - \bar{x})^2}}$$

; k = Coefficient Number, n = Number of Sample, n-k = Degree of Freedom

$H_0: \hat{b} = 0$, $H_a: \hat{b} \neq 0$, $t = \frac{\hat{b} - 0}{S_b}$; t > Critical Value $\rightarrow$ Reject H_0 and/or Accept H_a

Test of Goodness of Fit $\rightarrow R^2$: Coefficient of Determination: Power of Independent variable to Dependent variable

$$R^2 = \frac{\text{Explained Variation}}{\text{Total Variation}} = \frac{\sum_{t=1}^n (\hat{y}_t - \bar{y})^2}{\sum_{t=1}^n (y_t - \bar{y})^2}$$

R^2 affected by Sample size (Less size big R^2), Independent variable (a greater number of Independent valuable more R^2)

Corrected Coefficient of Determination

$$\bar{R}^2 = R^2 - \frac{(k-1)}{(n-k)}(1 - R^2) ; k = \text{number of coefficients} + \text{intercept}, n = \text{number of samples}$$

Multiple Regression Analysis

F-statistic

$$F = \frac{\text{Explain Variation} / (k-1)}{\text{Unexplain Variation} / (n-k)} = \frac{R^2 / (k-1)}{(1 - R^2) / (n-k)}$$

$$H_0: b_1 = b_2 = b_3 = \dots = b_n = .$$

$$H_a: \quad b \quad \neq 0$$

; $F > \text{Critical Value} \rightarrow \text{Reject } H_o \text{ and/or Accept } H_a$

Problem $\rightarrow$ Multicollinearity = Independent Variable Correlation 2

Solve $\rightarrow$ Independent Variable Correlation $\hat{b}$
Dependent Independent

Problem $\rightarrow$ Heteroskedasticity = Variance of error term [Cross-section data]

Solve $\rightarrow$ Take Log Use weighted least square regression variable

Problem $\rightarrow$ Autocorrelation/Serial Correlation = Error term

Solve $\rightarrow$ / Model

Process of Regression Analysis

1. Specify Variable $Q_x = f(P_x, P_s, P_c \dots \text{etc})$

2. Collecting Data on Variable

- Time Series Data = Data in period of time

- Cross-section data = Data in a point of time

3. Specifying to Equation $Q_x = b_0 - b_1 P_x + b_2 P_s - b_3 P_c \dots$

Take log...

4. Interpret Statistic Results $\rightarrow t, F, R^2, \bar{R}^2$

5. Interpret Results of Model

Balance of payment ..

*** 5**

*

*digital money